Experiment 6

Exploration of Cloud Service Models (IaaS, PaaS, SaaS) Using a Free-Tier Cloud Platform

1. Aim

To explore the differences between IaaS, PaaS, and SaaS and identify suitable use cases for each cloud service model.

2. Objectives

- Define Infrastructure as a Service, Platform as a Service, and Software as a Service.
- Compare user and provider responsibilities in each model.
- Explore a free-tier cloud console or representative service documentation.
- Select a suitable model for a given application requirement.

3. Requirements

Internet access, a cloud provider account if available, and awareness of free-tier limits and billing settings.

4. Theory

Cloud computing provides computing resources over a network. IaaS offers virtual machines, storage, and networking while the customer manages the operating system and applications. PaaS provides a managed environment for developing and deploying applications, reducing infrastructure administration. SaaS delivers a complete application through a browser or client, with most technical management handled by the provider. Responsibility varies by service, so the exact boundary should be checked in the provider's documentation. Free tiers may have quotas, regional limits, or charges for resources beyond the allowance.

5. Procedure

1. Choose a reputable cloud provider and review its free-tier terms before creating resources.
2. Explore the console categories for virtual machines, managed application hosting, databases, and ready-to-use software.
3. Classify one example as IaaS, one as PaaS, and one as SaaS.
4. Record what the provider manages and what the customer must configure.
5. If practical, create a small permitted resource and confirm how to stop or delete it.
6. Review usage dashboards, budgets, and billing alerts; remove test resources after the exercise.

6. Case Study and Analysis

A student team wants to create a web-based inventory application for a college laboratory. The team considers three approaches. With IaaS, it could rent a virtual machine and configure the operating system, web server, database, security updates, and application. This offers flexibility but requires more administration. With PaaS, the team could deploy code to a managed application platform and use a managed database. The provider handles more infrastructure tasks, allowing the team to focus on features. With SaaS, the laboratory could use an existing inventory product with minimal development, but customization may be limited. The best choice depends on the team's priorities: IaaS is useful when system-level control is necessary, PaaS suits custom application development, and SaaS is attractive when a ready-made solution meets the need. During a free-tier experiment, the team should check quotas and delete unused resources because free-tier availability does not guarantee that every service or usage level is free.

7. Result / Discussion

The comparison showed that IaaS, PaaS, and SaaS differ mainly in the level of management and customization provided to the customer.

8. Conclusion

Choosing a cloud model requires balancing control, development effort, operational responsibility, and cost. Free-tier exploration should be performed carefully with usage and billing controls.

9. Precautions / Best Practices

- Use fictional or non-sensitive test data during practice.
- Verify settings and results before using the application or configuration in a real environment.
- Record observations clearly and follow the platform's current documentation, security, and access-control guidance.